

Instructions for *ACL Proceedings

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Abstract

Soubki and Heinz (2023) examine the performance of the state-merging algorithms RPNI, EDSM, and ALERGIA were tested on an early version of MLRegTest, a benchmark for the learning of regular languages (van der Poel et al., 2024). They found that the neural models generally outperformed the state-merging algorithms on this benchmark. The MLRegTest benchmark excludes strings less than length 19 in order to achieve a balance of positive and negative strings in training, development and test sets across all languages in MLRegTest. Soubki and Heinz (2023) report a follow-up experiment which includes shorter strings and shows they dramatically improve the performance of the state-merging algorithms.

This paper makes three contributions. First, it replicates the experiments in Soubki and Heinz (2023) for the neural networks and state-merging algorithms RPNI and EDSM on the final version of the MLRegTest. This replication includes a grid search over many parameters belonging to the state-merging algorithms provided by the flexfringe software (Verwer and Hammerschmidt, 2017, 2022) to find suitable hyperparameters. Second, it conducts a comprehensive set of follow-experiments with shorter strings on *both* the neural and state-merging learning systems to allow for comparison. Third, on the basis of these experiments, we conclude that the neural systems outperform the state-merging systems in all conditions save one. When the training data is small, and the short strings are included, EDSM outperforms the neural networks.

1 Introduction

2 MLRegTest

3 Learning Systems

3.1 State-Merging Systems

3.2 Neural Systems

4 Experiments

5 Results

6 Discussion

7 Conclusions

References

Adil Soubki and Jeffrey Heinz. 2023. [Benchmarking state-merging algorithms for learning regular languages](#). In *Proceedings of 16th edition of the International Conference on Grammatical Inference*, volume 217 of *Proceedings of Machine Learning Research*, pages 181–198. PMLR.

Sam van der Poel, Dakotah Lambert, Kalina Kostyszyn, Tiantian Gao, Rahul Verma, Derek Andersen, Joanne Chau, Emily Peterson, Cody St. Clair, Paul Fodor, Chihiro Shibata, and Jeffrey Heinz. 2024. [MLregtest: A benchmark for the machine learning of regular languages](#). *Journal of Machine Learning Research*, 25(283):1–45.

Sicco Verwer and Christian Hammerschmidt. 2022. [FlexFringe: Modeling software behavior by learning probabilistic automata](#). *arXiv preprint*.

Sicco Verwer and Christian A. Hammerschmidt. 2017. [FlexFringe: A passive automaton learning package](#). In *2017 IEEE International Conference on Software Maintenance and Evolution (ICSME)*, pages 638–642.

A Example Appendix

This is an appendix.